

Bachelor of Technology (Computer Science and Engineering) Eight semester Major Project



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Submitted to:

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(HOD CSE)

Density-Based Traffic Controlling System :-

A Density based traffic controlling system aims to optimize traffic flow and reduce congestion by analyzing real time-traffic density .





Content



Real-time Data Collection

The system utilizes sensors to gather real-time traffic data, including vehicle density, speed, and flow.



Dynamic Traffic Signal Control

Traffic signals adapt to varying traffic conditions, optimizing signal timings to reduce congestion.



Adaptive Routing

Drivers receive real-time route guidance to avoid congested areas and minimize travel time.

Objective :-

1

Reduce Congestion

Primarily focuses on reducing traffic congestion in city .

2

Improve Traffic Flow

Optimize signal timings for smoother and more efficient traffic flow.

3

Enhance Safety

Enhance driver and pedestrian safety by reducing traffic-related accidents.



Proposed System



Sensors

Collect data on traffic density, speed, and flow.



Communication Network

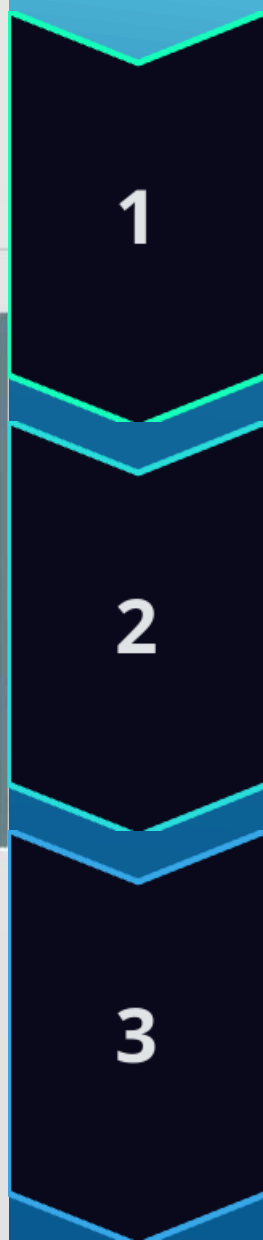
Transmits real-time data to the central control unit.



Central Control Unit

Processes data and dynamically adjusts traffic signals.

Analysis



- ## Traffic Flow Optimization

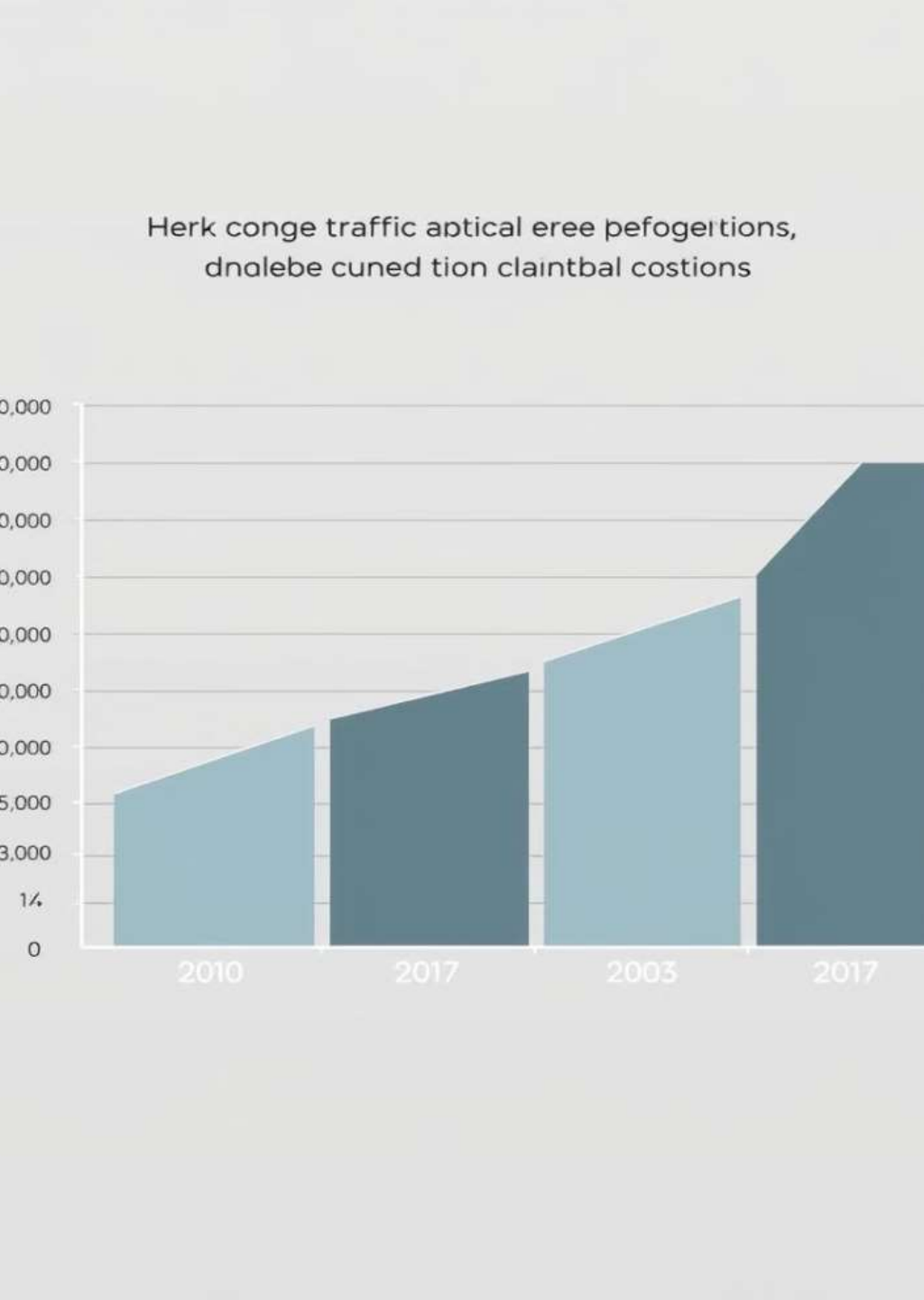
The system can significantly improve traffic flow efficiency by reducing bottlenecks and delays.
- ## Fuel Consumption Reduction

Smoother traffic flow can decrease fuel consumption and reduce greenhouse gas emissions.
- ## Accident Prevention

The system can enhance road safety by reducing congestion and improving driver behavior.

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Year	Traffic (approx.)	Costs (approx.)
2010	5,000	5,000
2017	7,000	7,000
2003	8,000	8,000
2017	10,000	10,000



1

2

3

Analysis

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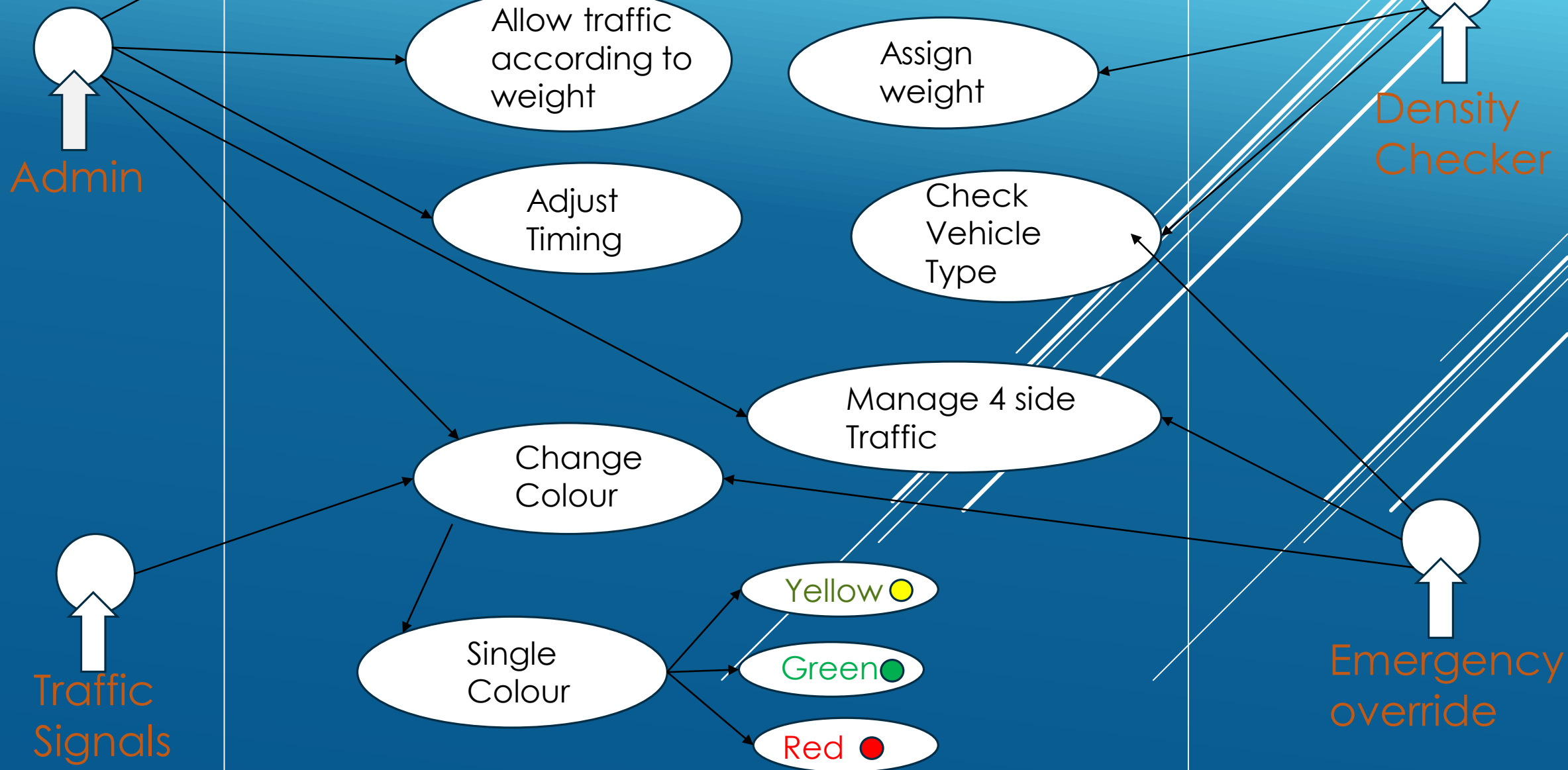
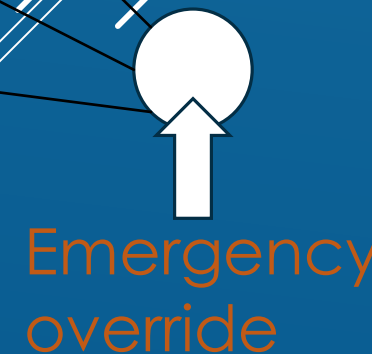
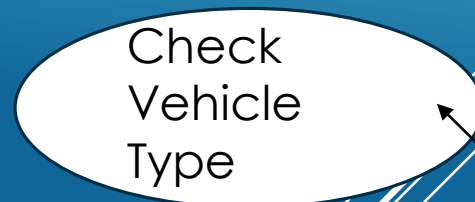
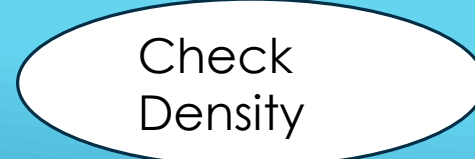
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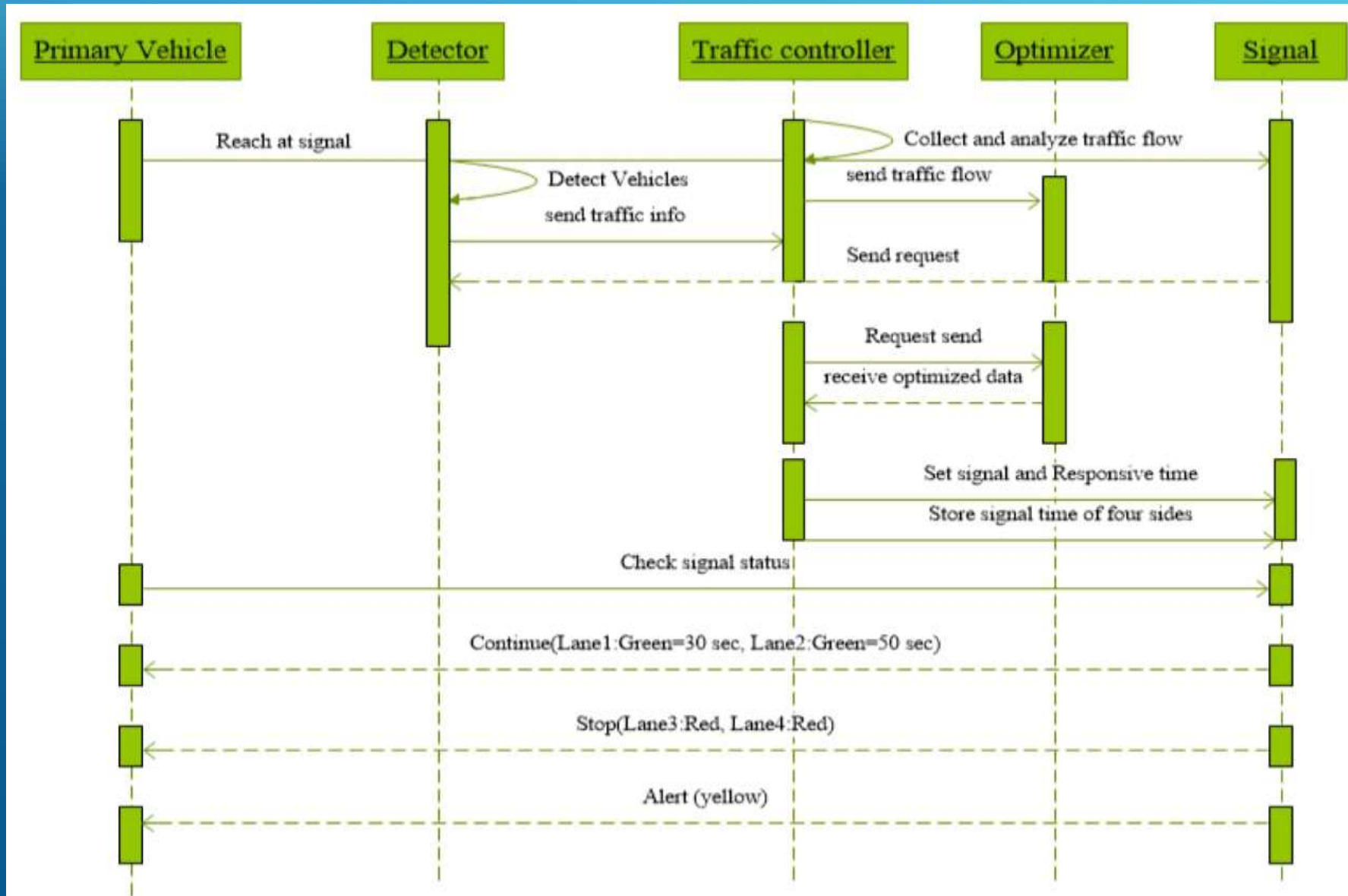
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Design



SEQUENCE DIAGRAM :-



Tools and Technology Used

1

Hardware Components

Micro Controller, Arduino, Camera, IR Sensor, Traffic Signals,
Communication Infrastructure

2

Software

Programming language: C++.

Cloud storage

3

Other Technologies

(IoT) for integrating various devices and sensors into system





Need and Significance

Traffic congestion poses a significant challenge to modern cities, impacting the economy, environment, and quality of life.

This system offers a promising solution by leveraging advanced technology to optimize traffic flow and reduce congestion.

Hypothesis and Expected Outcome

Hypothesis

We hypothesize that the density-based traffic control system will significantly reduce traffic congestion and improve travel times.

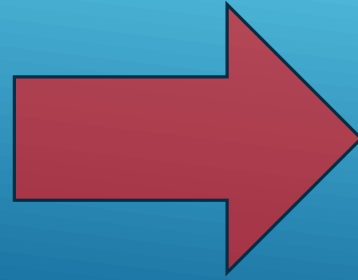
Expected Outcomes

We expect to see a substantial reduction in average travel times, improved traffic flow, and a decrease in vehicle emissions.

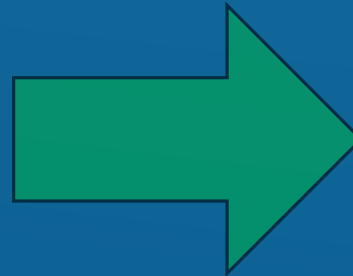




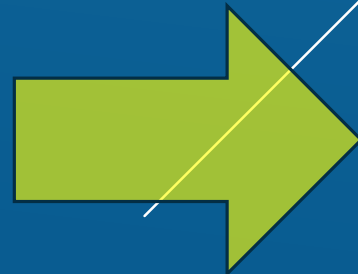
Future Scope



Real Time Traffic Information
Update On Traffic Department
Website



Automatic Tracking Of Traffic
Laws Defaulters



Implementation Of Above
System At Night using There
Cameras

Conclusion :-

The density-based traffic controlling system is an intelligent and adaptive solution for managing traffic flow in urban areas. By using real-time traffic density data, the system can optimize traffic signal timings to reduce congestion, decrease travel times, and improve overall traffic efficiency.

Thank you

The background is a blue gradient, transitioning from a lighter blue at the top to a darker blue at the bottom. Several thin, white diagonal lines are scattered across the right side of the image, creating a sense of motion or a modern design element.